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Lithium-Ion Conducting Electrolyte Salts for Lithium Batteries

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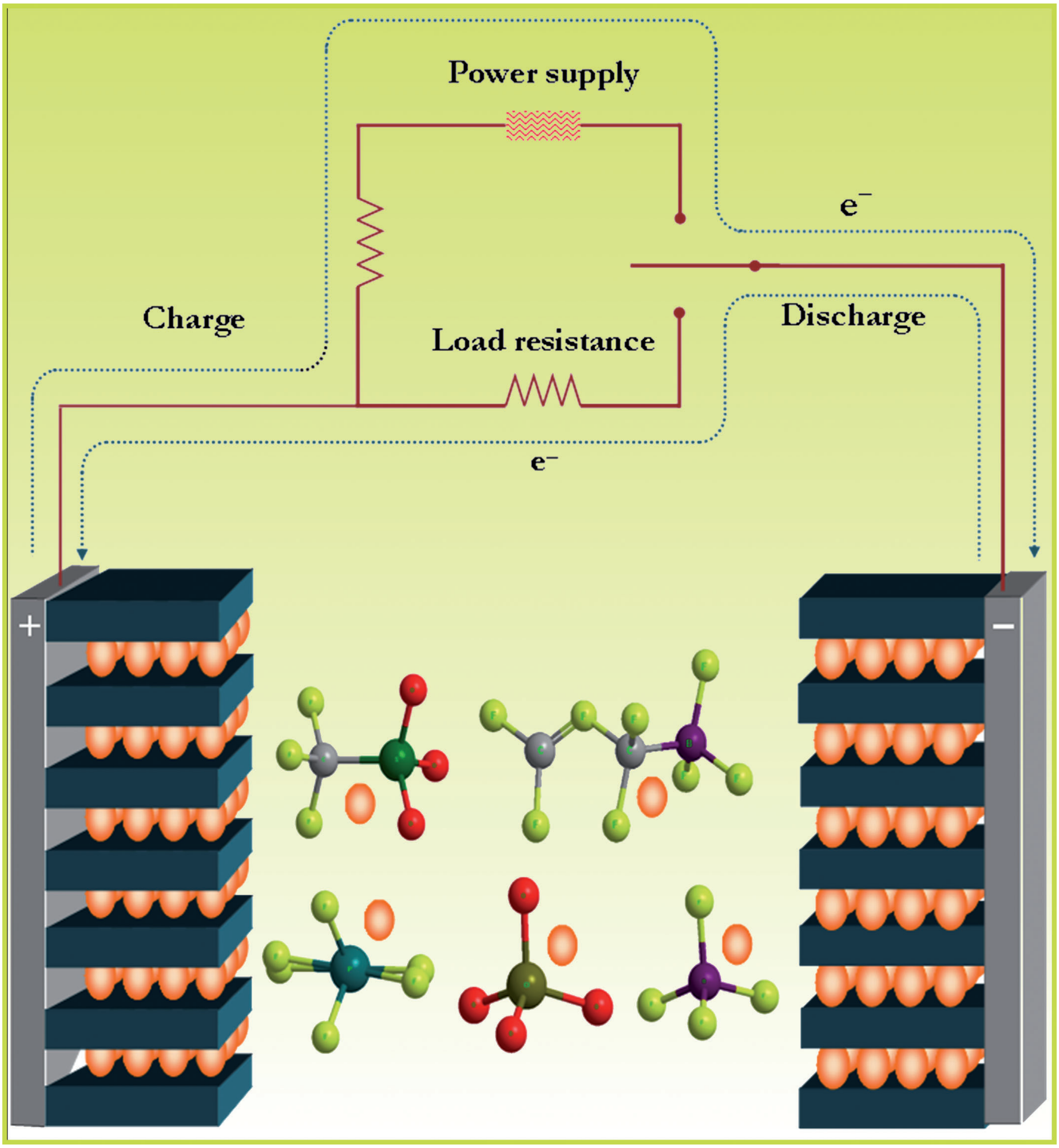

Abstract: This paper presents an overview of the various types of lithium salts used to conduct Li^+ ions in electrolyte solutions for lithium rechargeable batteries. More emphasis is paid towards lithium salts and their ionic conductivity in conventional solutions, solid–electrolyte interface (SEI) formation towards carbonaceous anodes and the effect of anions on the aluminium current collector. The physicochemical and functional parameters relevant to electrochemical properties, that is, electrochemical stabilities, are also presented. The new types of lithium salts, such as the bis(oxalato)borate (LiBOB), oxalyldifluoroborate (LiODFB) and fluoroalkylphosphate (LiFAP), are described in detail with their appropriate synthesis procedures, possible decomposition mechanism for SEI formation and prospect of using them in future generation lithium-ion batteries. Finally, the state-of-the-art of the system is given and some interesting strategies for the future developments are illustrated.

Keywords: electrochemistry • conducting materials • lithium • lithium-ion batteries

Introduction

In chemistry and physics, substances that conduct electric current do so as a result of dissociation into ions, which migrate toward and are discharged at the negative and positive terminals (cathode and anode) of an electric circuit. They can be divided into acids, bases and salts, because they all give ions when dissolved in water or organic molecules. These solutions conduct electricity due to the mobility of these cations and anions. Generally, the electrolytes are an indispensable constituent in all electrochemical devices. Their basic function is generally independent of the diverse chemistries of these devices; that is, the function of electrolytes in electrolytic cells, capacitors, fuel cells or batteries remains the same. The vast majority of the electrolytes are electrolytic solution-types that consist of salts dissolved in solvents, either water (aqueous) or organic molecules (non-aqueous), and are in a liquid state over the functional temperature ranges. When electrolytes are placed between the pair of electrodes and a voltage is applied, the electrolyte will conduct electricity by means of the transportation of charge carriers. Lone electrons normally cannot pass through the electrolyte; instead, a chemical reaction occurs at the cathode consuming electrons from the anode, and another reaction occurs at the anode producing electrons to be taken up by the cathode. As a result, a negative charge cloud develops in the electrolyte around the cathode and a positive charge develops around the anode. Consequently, the electrolyte should be inert towards the electrodes during the Faradaic reaction. Further, it should be a poor electronic conductor, but a good ionic conductor. The ions in the electrolyte move to neutralize these charges so that the reactions can continue and the electrons can keep flowing in the external circuit.

In typical rechargeable batteries, two electrodes with different electron affinities are used; electrons flow from one electrode to the other outside of the battery, while inside the battery the circuit is closed by the electrolyte's ions. Here the electrode reactions convert chemical energy to electrical energy. In a typical lithium-ion battery, Li^+ ions are shuttle between two lithium intercalation/insertion electrodes.[1–25] Lithium ions move from the cathode to anode during charge and back during discharge (Figure 1). The

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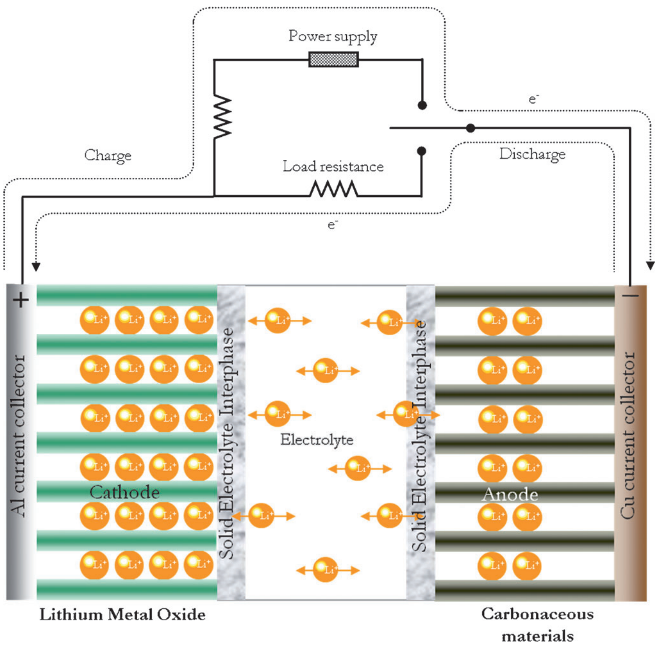

Figure 1. Working principal of a typical Li-ion cell.

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common candidates for the cathodes are lithiated metal oxides, while carbonaceous materials are the most popular anode material so far. This reversible transfer of lithium ions to and from interstitial sites in the host materials is accompanied by charge transfer. Ionic transport within the cell is ensured by an organic electrolyte, usually a solution of $LiPF_6$ in high-permittivity, low-viscous solvent mixtures made from the mixtures of cyclic and linear carbonates. Lithium-ion batteries do not employ metallic lithium as the anode. The lithium-ion battery is, therefore, often described as a metal-free lithium battery. The key to a safe and high-performance lithium-ion cell lies in the identification of a suitable electrolyte. It is known that lithium is intrinsically unstable with any commonly known electrolyte. So far, lithium salts studied in lithium battery electrolytes applications are very limited when compared to the electrodes as well as solvent combinations. Although solvents are quite harmless (non-toxic) and optimized for performance (carbonates mixtures, additives and so on), the lithium conducting salts still lacks many desirable properties. At the same time, their improvement is said to be the most prospective way to enhance and improve lithium cell parameters. However, commercial electrolyte solutions containing $LiPF_6$ possess some properties desirable for lithium-ion battery environment, so that replacing the salt will entail certain trade-offs. Nevertheless, today it is the best compromise, as alternatives to $LiPF_6$ pose too many problems (will be discussed in upcoming sections), leading to the possible explosion of the cell. In this review, we address an extensive analysis of the salts used in liquid electrolytes. Recently reported, new types of lithium-ion conducting salts, such as lithium bis(oxalato)borate (LiBOB), lithium oxalyldifluoroborate (LiODFB) and lithium fluoroalkylphosphate (LiFAP), are also discussed, from their synthesis to complete electrochemical properties, such as, ionic conductivity, electrochemical stability, cycleability, solid–electrolyte interface formation, over charge tolerance and so forth with appropriate mechanisms. [Note: The solid–electrolyte interface is a passivation film formed by reaction of the electrode material with the electrolyte. This thin layer (15–25 Å thickness) contains some insoluble products, which are electronically insulating, but it is a good conductor of lithium ions. It prevents direct contact between the electrode and the electrolyte, and suppresses further reaction between them.] The commercially available simple inorganic salts ($LiBF_4$, $LiClO_4$, $LiAsF_6$ etc.), salts containing bulkier anionic groups, such as triflates, methides and imides, are also compared, but more details about their applications in batteries are described elsewhere.[26] Other types of electrolytes (ceramic and solid polymer electrolytes, which include composite polymer systems) used for lithium batteries and additives for electrolytes are not included in this discussion. Before entering deeply into this topic, the new or existing salts should meet the following prerequisites to be used as electrolyte salts in lithium-ion battery electrolytes,

1) High thermal and chemical stability.
2) It should able to passivate an aluminium (Al) current collector from anodic dissolution.
3) It should exhibit high ionic conductivity in various non-aqueous solvent systems.
4) The anion should be stable against oxidative decomposition at the cathode.
5) It should be able to form less-resistive solid–electrolyte interfaceson electrodes (especially carbonaceous anodes) for cell safety and long-term cyclability.
6) It should be able to completely dissolve and dissociate in the non-aqueous media (wide variety of organic solvents, especially linear and cyclic carbonates), and the solvated ions (especially lithium cations) should be able to move in the media with high mobility.
7) Both anion and cation should remain inert towards the other cell components, such as the separator, electrode substrates and so forth.
8) The anion should be inert to electrolyte solvents.
9) The anion should be non-toxic and remain stable against thermally induced reactions with electrolyte solvents and other cell components.

Lithium Perchlorate ($LiClO_4$)

$LiClO_4$ is a salt of a strong acid, and has high solubility, good ionic conductivity in non-aqueous solvents and high anodic stability (up to 5.1 V on a spinel cathode surface in ethylene carbonate/dimethyl carbonate). $LiClO_4$ is stable towards hydrolysis compared the fluoro-type salts, $LiBF_4$ and $LiPF_6$, and there is no possibility of forming the HF in the electrolytes. Further, $LiClO_4$ is strongly involved in the formation of surface films on both carbonaceous and lithium metal anodes and exhibits good ionic conductivity around $9\,mS\,cm^{-1}$ in ethylene carbonate or dimethyl carbonate at room-temperature.[27]

The high oxidation state of chlorine (VII) makes the perchlorate a strong oxidant, which readily reacts with organic species at higher temperature, as well leading to high current charging; it may be explosive in above-mentioned environments.[28] For this reason, this salt cannot be used in practical batteries, because of its explosive nature;[29] however, it is one of the best candidates being used in various laboratories for testing electrolytes owing to its low cost, easy handling and its greater stability relative to fluoro-type salts. The electrochemical stability of $LiClO_4$ has been measured, and it suggests that it should not be used in more than 4.5 V (vs. Li/Li^+), otherwise it leads to the decomposition of ClO_4^- ions in to ClO_2 and HCl.[30] $LiClO_4$ has low interfacial resistance among the salts discussed, which is due to that fact that it does not form thin resistive LiF-like species.[31]

The conductivity of $LiClO_4$-based polymer electrolytes mainly depends on the temperature and concentration of the salt, suggesting that transfer of conductivity not only by the simple ions Li^+ and ClO_4^-, but also by ion clusters like $Li(ClO_4)_2^-$ and $Li_2(ClO_4)^+$,[32] and anions of $LiClO_4$ are effectively preventing the polymer chain reorganization.[33] $LiClO_4$ decomposes during the operation of the cell and decomposition mechanism are given in Equation (1).[34,35]

$$LiClO_4 + ne^- + 2nLi^+ \rightleftharpoons Li_2O + LiClO_3,\ LiClO_2 + LiCl \qquad (1)$$

The byproduct from this reaction, $LiClO_3$, is highly hygroscopic in nature; it absorbs the moisture and leads to the explosion of the cell during cycling.[36]

Lithium Tetrafluoroborate ($LiBF_4$)

$LiBF_4$ has been widely studied as it has a smaller anionic size (0.227 nm) relative to $LiPF_6$ (0.255 nm). Decreasing the anionic size increases the ionic conductivity due to the higher mobility of the BF_4^- ions relative to PF_6^-.

$LiBF_4$ is affected by high association with the Li^+ ion, owing to the smaller size of the anion (0.227 nm); the ionic contact distance is short, which leads to closer coordination with cations compared to $LiPF_6$ (0.255 nm).[37] The ionic contact distance increases with increasing size of the anion, leading to weaker ionic coordination with cations and improved conductivity. Weaker coordination indicates higher dissociation and also higher viscosity and ion mobility. The orders of ion mobility for some commonly available salts are as follows: $LiBF_4 > LiClO_4 > LiPF_6 > LiAsF_6 > LiTf >$ LiTFSI (Tf = trifluormethansulfonate; TFSI = bis(trifluoromethanesulfonyl)imide)

$LiBF_4$ is affected by the moisture problem like $LiPF_6$, even with only a trace amount of absorbed moisture it undergoes hydrolysis and the end product will be HF, which destroys the transition-metal element present in the cathode as well as Li^+ ion. The simplest hydrolysis reaction for $LiBF_4$ is given in Equation (2).

$$Li^+ + BF_4^- \leftrightarrow LiF\downarrow + BF_3\uparrow \qquad (2)$$

The BF_3 is highly reactive Lewis acid, and reacts violently with bases; it also readily reacts with moisture and forms dangerous HF as the end product [Eq. (3)].[38]

$$BF_3 + H_2O \rightarrow BOF\uparrow + 2\,HF \tag{3}$$

BF_3 also reacts with other functional groups, like decomposition products of solvent molecules (ethylene carbonate based solvents). The possible reactions are given in Equations (4)–(6).

$$BF_3 + Li_2CO_3 \rightarrow 2\,LiF\downarrow + BOF\uparrow + CO_2 \tag{4}$$

$$BF_3 + (CH_2OCO_2Li)_2 \rightarrow 2\,LiF\downarrow + BOF\uparrow + CO_2 + C_2H_2\uparrow + {}^1\!/_2O_2\uparrow \tag{5}$$

$$BF_3 + 2\,CH_2OCO_2Li \rightarrow 2\,LiF\downarrow + BOF\uparrow + CO_2 + CH_3O^{\bullet} \tag{6}$$

The absence of Li_2CO_3 in the decomposition products and the fact that trace amounts of HF are often present suggests that the reaction shown in Equation (7) may also play a part in the formation of thin LiF.

$$2\,HF + Li_2CO_3 \rightarrow 2\,LiF\downarrow + H_2O + CO_2 \tag{7}$$

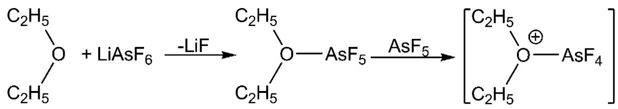
LiF is the main component of the solid–electrolyte interface formed from an $LiBF_4$-based electrolyte; no carbonate-like species are observed—these species are readily removed in the presence of $LiBF_4$. The formation of polymeric species and alkyl carbonate reduction components (formed from the decomposition of ethylene carbonate based solvents) can also be observed. The solid–electrolyte interface covers the entire anode surface and prevent unwanted reactions with the anode; however, this particular interface is not stable (beyond room temperature), like Li–imide-based electrolyte, but it is better than $LiPF_6$-based electrolyte.[39]

Moderate ionic conductivity (σ) is observed in $LiBF_4$-based electrolytes ($\sigma \approx 4.9\ mS\,cm^{-1}$) in ethylene carbonate/dimethyl carbonate (30:70),[40] which is inferior to that for $LiPF_6$ ($\sigma \approx 10.7\ mS\,cm^{-1}$) in ethylene carbonate/dimethyl carbonate (30:70);[41] however, it is thermally more stable than the commercially used $LiPF_6$ salt.[42] $LiBF_4$ passivates the Al current collector and a solid–electrolyte interface is formed on both the carbonaceous anodes and the cathodes. The passivation phenomenon prevents the dissolution or desorption of the Al current collector. BF_4^- is soft anion that reacts with the Al current collector and forms species like AlF_3 and $Al_2(CO_3)_3$, which hinder the desorption of Al and $Al(BF_4)_3$) from the surface of the current collector. From this, it is concluded that $LiBF_4$ does not react more strongly with Al current collector than other salts like $LiPF_6$, $Li(CF_3SO_3)$, $LiC(CF_3SO_2)_3$ and $Li[N(CF_3SO_2)_2]$. The anodic current is higher in the case of $LiBF_4$ against carbonaceous anodes, which indicates the formation of a stable solid–electrolyte interface at ambient temperature, but it is unstable at elevated temperatures. This leads to the self-discharge and severe capacity fading of the cell during the cycling.[37] Further, the main setbacks of using $LiBF_4$ are 1) inferior ability, relative to that of $LiPF_6$, in assisting the formation of solid–electrolyte interface on the surface of graphite electrode and 2) poor ability in forming a supercooled solution of the electrolytes at low temperatures, either because of its low solubility or because of freezing of the electrolytic solvents. The former is due to the fact that the BF_4^- ion is too stable to participate in the solid–electrolyte interface formation, and the latter is due to the high symmetry and small size of the BF_4^- ion. Reduction in the stability and symmetry of the BF_4^- ion is assumed to alleviate the drawbacks of $LiBF_4$, which have restricted its application in the electrolyte of Li-ion batteries.

Lithium Hexafluoroarsenate ($LiAsF_6$)

$LiAsF_6$ has been tested in many laboratories for its use in electrolyte salts for lithium batteries. It exhibits ionic conductivity (σ) slightly higher ($\sigma \approx 11.1\ mS\,cm^{-1}$) than $LiPF_6$ ($\sigma \approx 10.7\ mS\,cm^{-1}$) in ethylene carbonate/dimethyl carbonate (1:1).[41] It was found that $LiAsF_6$ is a superior salt to $LiClO_4$; it is even soluble in low-dielectric-constant solvents (diethyl carbonate, dimethyl carbonate etc.) and has an average cycling efficiency is $>95\,\%$.[43] $LiAsF_6$ robustly passivates the Al current collector, and forms a stable solid–electrolyte interface on the both electrode surfaces; these surface layers are similar to those of $LiBF_4$- and $LiPF_6$-based salts, nevertheless the species mainly consists of alkyl carbonates (Li_2CO_3) rather than LiF.[44,45–47]

The strong As–F bonds are less labile compared to the P–F bonds in $LiPF_6$, which results in the reduced formation of HF.[48] $LiAsF_6$ strongly reacts with ether-based solvents and produces gaseous byproducts as shown in Scheme 1.[49]



Scheme 1.

Further, AsF_6^- ions might react with ether solvents to give the following by-products: $CH_2{=}CH_{2(gas)}$, CH_3CH_2F, $CH_3CH_2AsF_6$, polymeric species, and so forth.

It is known that As^V is not toxic, but that As^{III} and As^0 are highly toxic in nature and the use of As in commercial batteries can lead to serious consequences for the users.[50] Anodic stability of the salt is found to be very high[50,51]—around 6.3 V vs. Li/Li^+—and experimental values around 4.5 V for various cathodic surfaces make this a promising candidate for electrolyte salts used in high-energy batteries but the toxicity of the arsenate remains a major concern.[52] The stability of the anions is given as follows: $CF_3SO_3^- < ClO_4^- < (CF_3SO_2)_2N^- < (CF_3SO_2)_3C^- < C_4F_9SO_3^- < BF_4^- < PF_6^- < AsF_6^-$; the strong anodic stability is noted, which is due to the strong nature of the acidity.[53]

Moisture is also one of the main problems in the case of $LiAsF_6$, as it leads to the production of dangerous products as well as HF. Pure solvents do not change the nature of the surface film, but the strong Lewis acid nature of AsF_6^- ion will change the decomposition of the salt as shown in Equations (8)–(10).[34,54]

$$LiAsF_6 + H_2O \rightarrow HF + AsF_5 + LiOH \quad (8)$$

$$LiAsF_6 + 2e^- + 2Li^+ \rightarrow AsF_3 + 3LiF \quad (9)$$

$$AsF_3 + 2ne^- + 2nLi^+ \rightarrow Li_nAsF_{3-n} + nLi^+ \quad (10)$$

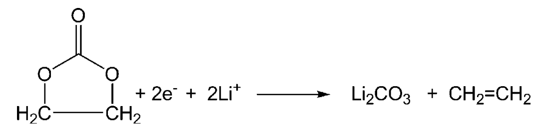
The dangerous products HF, AsF_5, or AsF_3 can react with the surface layers (SEI) and break them, resulting in the formation of very thin but robust film that is mainly composed of LiF. This property has not been observed in any of the salts studied and during prolonged cycling the surface film also grows, which hinders the intercalation and de-intercalation of Li^+ ions, and leads to the cell working in a thermal runaway.[55,56]

$LiAsF_6$ has some superior properties over $LiPF_6$ and it is also involved in the formation of surface films that are more flexible than those formed by $LiPF_6$ on both graphitizable (soft) and non-graphitizable (hard) carbonaceous anodes. Eshkenazi et al.[57] reported that more reduction products occur in $LiAsF_6$-based electrolytes, such as HF. As_xF_y and LiF are also found in the solid–electrolyte interface, while carbonates are not detected on soft carbons; this is due to hindrance of ethylene carbonate decomposition. The main components involved in the formation of the solid–electrolyte interface are LiF and As–F compounds (ca. 40%), polyolefins (ca. 20%), C–O groups (ca. 20%) and Li_2O (ca. 20%; Figure 2). The mechanism of As–F bond formation is mentioned above.

Li_2O content in the solid–electrolyte interface are accompanied by the higher concentration of carbonates (hard carbons/$LiAsF_6$ electrolytes), this type of secondary reaction was proposed by Yang et al.[58] and Arora et al. [Eq. (11)].[59]

$$CO_3^{2-} + 5e^- + 7Li \rightarrow 3Li_2O + 0.5Li_2C_2 \quad (11)$$

Carbonates are the sources of solvent reduction; for example, ethylene carbonate, which undergoes a one-electron decomposition (Scheme 2).[60]



Scheme 2.

The solid–electrolyte interface formed on hard carbon is different from that formed on soft carbon. The hard-carbon solid–electrolyte interface is mainly composed of carbonates, owing to the decomposition of ethylene carbonate in to ethylene and lithium carbonates. Hence, the carbonates are mainly reduced to Li_2O and Li_2C_2 which make up around 10% of the solid–electrolyte interface. In the case of hard carbons, LiF and As_xF_y compositions account for around 20% of the solid–electrolyte interface, which is about half that found for soft carbons. A high content of insoluble carbonate were found in the solid–electrolyte interface; this hinders the Li^+ intercalation, leads the thermal runaway, and possibly even to breaking the solid–electrolyte interface. An XPS study indicates the existence of C–H, C-O-C, C=O and Li-O-C groups, along with carbonate moieties. The possible compounds present in the solid–electrolyte interface are schematically given in Figure 3.

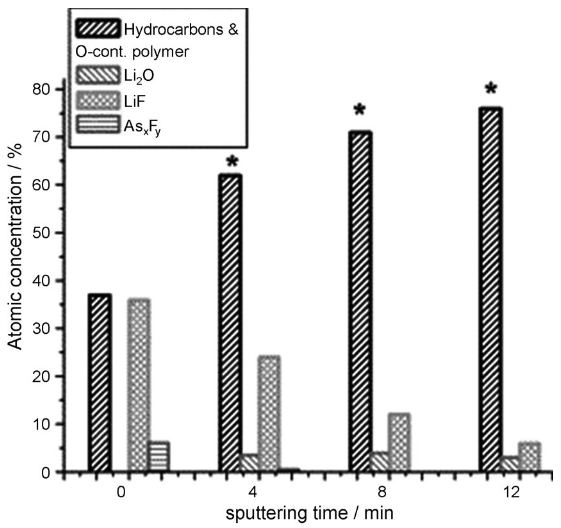


Figure 2. Proposed content of the SEI on the soft carbon in $LiAsF_6$ electrolyte. *: including substrate carbon. From reference [57]; reproduced with permission from Elsevier.

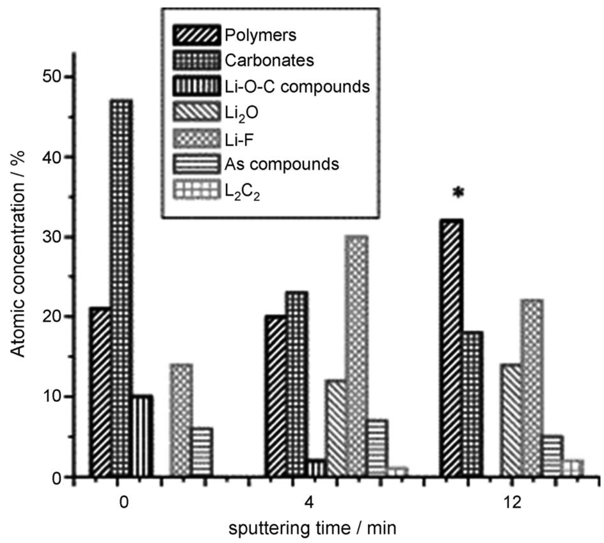


Figure 3. Proposed content of the SEI on the hard carbon in $LiAsF_6$ electrolyte. *: including substrate carbon. From reference [57]; reproduced with permission from Elsevier.

Lithium Hexafluorophosphate ($LiPF_6$)

$LiPF_6$ has been extensively studied by many researchers and although it has a serious moisture problem, it remains the only candidate to be widely used in commercial lithium-ion batteries. The anion of lithium hexafluorophosphate ($LiPF_6$) could be viewed as F^- complexed by Lewis acid PF_5. Such anions, also known as anions of super acids, have a structure in which the formal negative charge is well distributed by the strongly electron-withdrawing Lewis acid ligands and the corresponding complex salts usually have lower melting points and more soluble in low dielectric media than their parent salts. The unanimous choice for this salt is not purely based on the above-mentioned properties; it has only slightly lower conductivity than the $LiAsF_6$ [the conductivity of $LiPF_6$ is approximately 10.7 $mS\,cm^{-1}$ in ethylene carbonate/dimethyl carbonate (1:1), only fractionally lower than $LiAsF_6$ (ca. 11.1 $mS\,cm^{-1}$)].[45] $LiPF_6$ has a lower dissociation constant than LiTFSI (due to the strong coordination with the Li^+ ion relative to LiTFSI), and appreciable anionic mobility and lower mobility than $LiBF_4$ (due to the larger PF_6^- ionic size (0.255 nm) compared to BF_4^- ions (0.227 nm)).[61–63] It has slightly lower thermal and anodic stabilities compared to most of the other salts like $LiAsF_6$ and $LiClO_4$.[52] $LiPF_6$ has poor ambient moisture stability compared with $Li(CF_3SO_3)$ and $LiClO_4$ (because the P–F bond is rather liable toward hydrolysis even by trace amounts of moisture in non-aqueous solvents).

However, $LiPF_6$ compares well relative to other available salts based on the following properties.

1) Dissociation constant: $LiTf < LiBF_4 < LiClO_4 < LiPF_6 < LiAsF_6 < LiTFSI < LiBOB$
2) Average ion mobility: $LiBF_4 > LiClO_4 > LiPF_6 > LiAsF_6 > LiTf > LiTFSI$

At elevated temperature, operation is another problem due to the poor thermal stability; $LiPF_6$ exists in various forms, while absorbing the trace amount of H_2O (in ppm level) that exists in the electrolyte solution [Eqs. (12) and (13)].

$$LiPF_6 + H_2O_{(moisture)} \rightleftharpoons LiF + PF_5 \quad (12)$$

$$PF_6^- + 2\,e^- + 3\,Li \rightarrow 3\,LiF + PF_3 \quad (13)$$

PF_5 is a gaseous product and a strong Lewis acid; it initializes some desired/or undesired reactions (in the presence of non-aqueous solvents), such as ring opening, and absorbing moisture leads to the formation of HF. Thereby, it produces the polymeric species on the cathodic side that prevents the release of transition-metal elements (if the transition-metal elements are released, the battery will work on the thermal runaway).[54]

P–F bonds are highly susceptible to hydrolysis even at room-temperature to produce HF [Eq. (14)]. At elevated temperatures PF_5 tends to undergo hydrolysis, resulted the following dangerous byproducts [Eq. (15)].[64–67]

$$LiPF_6 + H_2O\ (\text{available in the electrolyte}) \rightarrow POF_3 + LiF + 2\,HF \quad (14)$$

$$PF_5 + H_2O \rightarrow POF_3 + 2\,HF \quad (15)$$

$LiPF_6$ is involved in the solid–electrolyte interface formation on both graphitizable (soft) and non-graphitizable (hard) carbon anodes. Eshkenazi et al.[57] briefly reported XPS studies of a solution containing $LiPF_6$/ethylene carbonate/dimethyl carbonate on soft carbon; the results showed that the solid–electrolyte interface mainly consists of LiF (around 50%) and polymeric species (around 30%)—the remaining part is composed of P_xF_y and LiOC-based compounds. A small amount of Li_2O is also found in the solid–electrolyte interface and some other possible solid–electrolyte interface products are given in the Figure 4; however, the solid–electrolyte interface layer is thinner compared to those formed in $LiAsF_6$-based electrolytes.

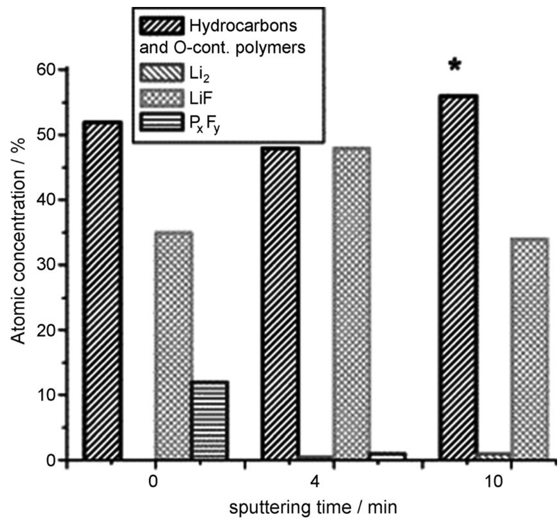


Figure 4. Proposed content of the SEI on the soft carbon in $LiPF_6$ electrolyte. *: including substrate carbon. From reference [57]; reproduced with permission from Elsevier.

On hard carbons, a different type of solid–electrolyte interface is formed, with an LiF content around 35–38%, and polymers and C–O-containing compounds are about 40–52%; the remaining PF_x compounds (ca. 12%) are formed according to the decomposition of the salt in ethylene carbonate. In this case a small amount of Li_2O is also detected; the composition of the solid–electrolyte interface products on hard carbon are given in Figure 5. The $Li_xPF_yO_z$ component is an electrolyte reduction product and cannot be neglected in both hard and soft carbon cases. In addition, no carbonate species are formed and it is believed that the reduction process of ethylene carbonate has been restricted in both cases of carbonaceous materials.

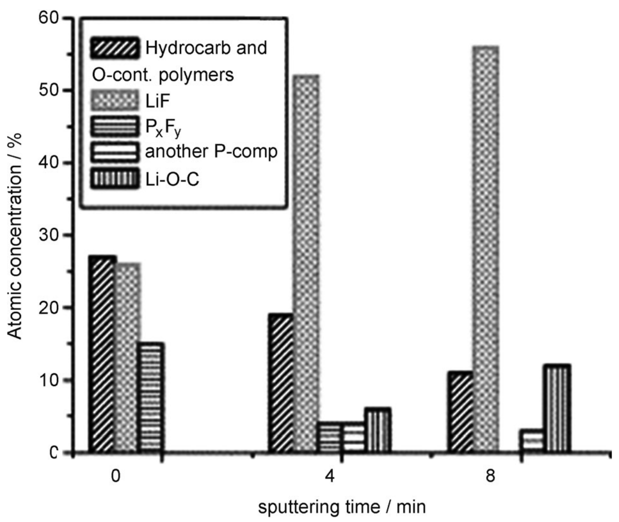


Figure 5. Proposed content of the SEI on the hard carbon in $LiPF_6$ electrolyte. *: including substrate carbon. From reference [57]; reproduced with permission from Elsevier.

LiF and PF/or P–O compounds form around 65 % of the solid–electrolyte interface on hard carbons, while on soft carbons they account for less than 50 %, owing to different catalytic properties that lead to different surface films. The solid–electrolyte interface on hard carbons is thicker than on soft carbons owing to co-intercalation of ion aggregates, like $Li_2PF_6^+$, at beginning of the surface film formation; this leads to the formation of $Li_xPF_yO_z$ or PO_3^-. However, the partial co-intercalation of protons from HF dissociation cannot be excluded. Partial exfoliation of hard carbons by $Li_2PF_6^+$ and penetration of hydrogen cause the increased thickness on the hard carbons.[68]

The anodic current is slightly less than for $LiBF_4$, which indicates that it forms on stable solid–electrolyte interface on both carbonaceous anodes and cathodes.[37] $LiPF_6$ also effectively passivates the Al current collector and prevents the dissolution or desorption of Al. Further, PF_6^- is a strong Lewis acid that reacts with the Al to form species like AlF_3, $Al_2(CO_3)_3$ (which hinders the desorption of Al), and $Al(PF_6)_3$. The reactivity of $LiPF_6$ towards the Al current collector is found to be much less than that found for $Li(CF_3SO_3)$. Currently, the two salts, lithium fluoroalkylphosphate and lithium bis(oxalato) borate, are competing to replace $LiPF_6$ in commercially available lithium-ion batteries.

Lithium Trifluoromethanesulfonate ($Li(CF_3SO_3)$)

Another family of lithium salts is based on the conjugate bases of the organic superacids, in which the acid strength is increased because of the stabilization of the anions by strongly electron-withdrawing groups, usually perfluorinated alkyls. In these anions, the delocalization of the formal negative charge is practically realized by a combination of the inductive effect of the electron-withdrawing groups and the conjugated structures. The attempt to use these salts originated from the hope that their dissociation constants would be high, even in low dielectric media, and that the organic nature of perfluorinated alkyls would assist the solubility of these salts in non-aqueous solvents. $Li(CF_3SO_3)$ is an impressive salt, which has been widely studied, but it exhibits poor ionic conductivity in non-aqueous medium. $Li(CF_3SO_3)$ is a superacid, as the acid strength is increased by the fluoroalkyl groups. These strong electron-withdrawing groups ($[CF_3SO_3]^-$) are hydrophobic in nature and hence resist hydrolysis. The delocalization of the negative charge on the $Li(CF_3SO_3)$-based anions results from the combined inductive effects of the electron-withdrawing groups.

It is believed that the formation of ion pairs, even in solvents with low dielectric constants, causes poor ionic conductivity. Such ion-pair formation was confirmed by Kim et al.[69] by using the FT-IR studies. For example, to increase the polymer concentration (poly(methyl methacrylate)), a decrease in either the carrier concentration or the ionic mobility is observed. Ionic distribution was analyzed for the stretching mode of SO_3 in triflate anions ($[CF_3SO_3]^-$) in polymeric gel electrolytes. The non-degenerative vibrational modes of $\nu_s(SO_3)$ appearing at 1030–1034, 1040–1045, and 1049–1053 cm^{-1} are assigned to free triflate anions, monodendate, ion-paired triflates (LiX, LiX_2^-, LiX_3^{2-}) and highly aggregated triflates (Li_2X^+ and Li_3X^{2+}) respectively.[70–73]

Generally, organic lithium salts that do not contain fluorine atoms and do not have a resonance structure showed poor solubility in organic solvents, even though they belong to the family of superacids. However, $Li(CF_3SO_3)$ has comparable anodic stability (4.8 V vs. platinum electrode while propylene carbonate as the solvent) and exhibited poor conductivity (ca. 1.7 $mS\,cm^{-1}$) in propylene carbonate based electrolytes. The choice of sulfonate group ($-SO_3Li$) in $Li(CF_3SO_3)$ reflects good thermal stability; it is also non-toxic, highly resistant towards oxidation and insensitive to ambient moisture conditions comparable to $LiPF_6$ and $LiBF_4$.[51] However, a main problem is severe corrosion towards Al current collector and it strongly reacts with Al yielding aluminium triflate ($Al(CF_3SO_3)_3$) formed as a by-product. The non-resistive properties of the $Al(CF_3SO_3)_3$ film leads to dissolution of current collector. Addition of HF into a solution containing $Li(CF_3SO_3)$ solution slightly inhibits the reactivity towards Al. However, surface species like Al_2O_3 and AlF_3 are formed due to the Lewis acid/base interaction between the HF and Tf^- and the above said species that inhibit the Al corrosion. When the voltage is increased to >3.6 V, the passivation film is destroyed and severe corrosion takes place. Further, addition of HF depresses the corrosion behaviour to a certain level, from 3.6 to 3.8 V; nevertheless, addition of HF is not recommended because of its disadvantageous properties.[74] Due to the above-mentioned facts this $Li(CF_3SO_3)$ is not used in practical lithium ion cells. The reduction mechanism of the $Li(CF_3SO_3)$ is shown in Equations (16)–(18)].[75]

$$2\,Li(CF_3SO_3) + 2\,e^- + 2\,Li^+ \rightarrow 2\,Li_2SO_3 \downarrow + C_2F_6 \uparrow \quad (16)$$

$$C_2F_6 + 2\,e^- + 2\,Li^+ \rightarrow CF_3CF_2Li + LiF \downarrow \quad (17)$$

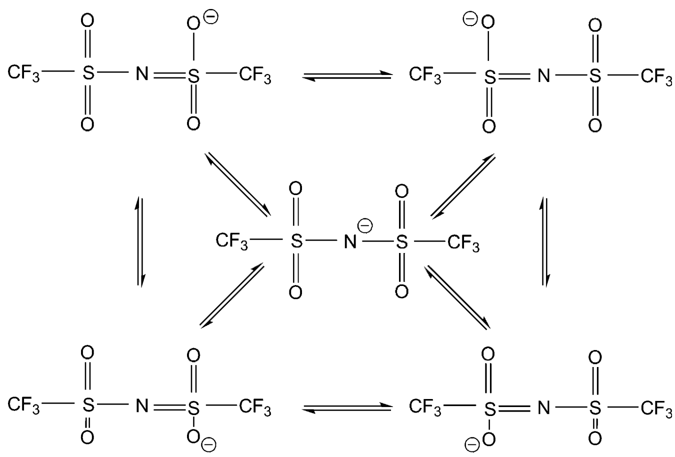
$$Li_2SO_3 + 6\,e^- + 6\,Li^+ \rightarrow Li_2S \downarrow + 3\,Li_2O \downarrow \quad (18)$$

The reduction process involves the 6+2+2 electrons and does not correspond to the irreversible capacity loss. This is observed during the first charge when triflate is used. Some excellent work was carried out by Kita et al.;[76] by increasing the chain length of alkyl groups (e.g., C_4F_9 and C_8F_{17} instead of CF_3) improvements were observed in the anodic stability ($Li(CF_3SO_3)$ −4.8 V, $Li(C_4F_9SO_3)$ −6.0 V and $Li(C_8F_{17}SO_3)$ −6.5 V in the propylene carbonate based solution vs. platinum electrode) and ionic conductivity ($Li(CF_3SO_3)$: ca. 0.4 $mS\,cm^{-1}$, $Li(C_4F_9SO_3)$: ca. 2.3 $mS\,cm^{-1}$ and $Li(C_8F_{17}SO_3)$: ca. 1.9 $mS\,cm^{-1}$ in the propylene carbonate/DME (1:2) based binary solutions). In $Li(C_8F_{17}SO_3)$-based electrolytes, the ionic conductivity is slightly decreased, because of conductivity mainly depends on the mobility of the dissociated ions in the solution. In this solution, the very large anion size decreases the mobility, hence the conductivity is decreased, but the values are better than $Li(CF_3SO_3)$-based solutions. The $Li(C_4F_9SO_3)$-based electrolyte also showed better performance among the three types of salts we have discussed in this section. The advantage of those salts can be described thus: Al corrosion is improved due to the larger anionic size, which efficiently passivates the surface film at the Al current collector.

Lithium Bis(trifluoromethanesulfonyl)imide ($Li[N(CF_3SO_2)_2]$ or LiTFSI)

In 1984, Foropoulos and DesMarteau[77] reported LiTFSI salt for use in rechargeable lithium batteries. Known as "lithium imide", the 3M Corporation has commercialised the salt for research and industrial purposes. The TFSI ion is stabilized by two trifluoromethanesulfonyl groups due to strong electron-withdrawing groups and a lone nitrogen atom; in addition the negative is well delocalized over the imide anion, as shown in Scheme 3.

During the period of commercialization, it was shown that LiTFSI salt is a promising alternative for lithium batteries and several investighations were reported within short period of time.[26] LiTFSI is thermally stable (melting temperature is 236 °C and it does not decompose until 360 °C) and highly conducting (but slightly lower conductivity than the $LiAsF_6$ and $LiPF_6$[45]). It also dissociates well in low-dielectric-constant solvents, leading to higher dissociation; the solution viscosity also increases (mobility tends to decrease), favouring higher ionic conductivity due to the larger anionic size.[51] The oxidation potential of LiTFSI is found to be 4.3 V vs. Li/Li^+ in ethylene carbonate/diethyl carbonate (1:1) binary solutions; this value is very high compared to other salts studied, apart from $LiAsF_6$.[78]



Scheme 3.

The TFSI anion actively participates in the formation of surface films; the NSO_2^- group is strongly involved in such processes. The complete reduction process of the TFSI anion in THF solution is described in Equations (19)–(21).[79,80]

$$Li[N(CF_3SO_2)_2] + 4\,e^- + 4Li^+ \rightarrow Li_3N \downarrow + 2\,Li(CF_3SO_2) \downarrow \quad (19)$$

$$2\,Li(CF_3SO_2) + n\,e^- + n\,Li^+ \rightarrow Li_2S_2O_4 \downarrow + C_2F_xLi_y + LiF \downarrow \quad (20)$$

$$Li_2S_2O_4 + 6\,e^- + 6\,Li^+ \rightarrow 2\,Li_2S \downarrow + 4Li_2O \downarrow \quad (21)$$

Laik et al.[81] proposed the above processes involving the 4+6 electrons, whichdoes not corresponding to the irreversible capacity loss that is observed in the first charging of the cell when "lithium imide" is used as the electrolyte salt. Decomposition of alkyl carbonates into Li_2CO_3 and polymerization of solvents can form a thermally stable passivation film that covers entire surface of the carbonaceous anodes. Salt anions with active fluorine atoms ($LiPF_6$, $LiAsF_6$ and $LiBF_4$) are able to form surface films on the Al current collector and protect the same. LiF and AlF_3 are the crucial products to prevent the corrosion of the Al current collector.[74] So far LiTFSI has not been used in commercialised rechargeable lithium ion cells owing to the strong corrosion behaviour at the Al current collector.[82] Though, the LiTFSI salt is widely used in ionic liquids as a source of Li^+ ions, it strongly reacts with Al and produce by-products based on aluminium (Al(TFSI)) owing to the strong adsorption behaviour of $TFSI^-$ ions towards Al. Also Al(TFSI) is not able to form a protective film on Al surface, leading to the dissolution of Al in non-aqueous solvents.[83]

A possible Al corrosion mechanism is given in Equations (22) and (23)[84] and a schematic representation is given in Figure 6.

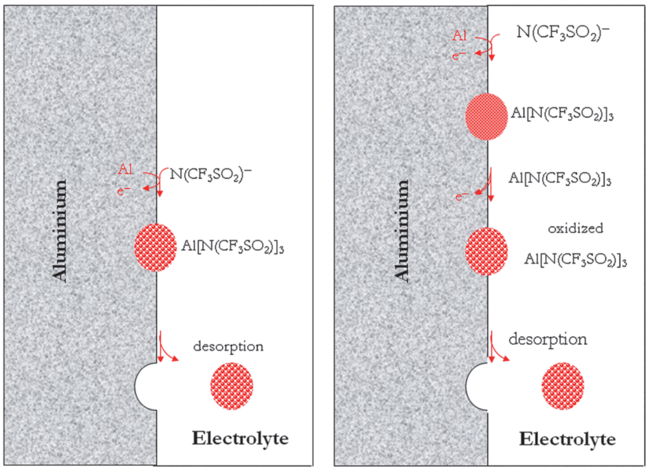


Figure 6. Two possible mechanisms for aluminium corrosion in $Li[N(CF_3SO_2)_2]$/propylene carbonate electrolytes.[84] Reproduced and modified with permission from Journal of the Electrochemical Society.

$$Al \xrightarrow{-3e} Al^{3+} \xrightarrow{X} AlX^{2+}{}_{(ad)} + AlX_2{}^{+}{}_{(ad)} + AlX_{3\,(ad)} \tag{22}$$

$$AlX^{2+}{}_{(ad)} + AlX_2{}^{+}{}_{(ad)} + AlX_{3\,(ad)} \underset{\text{re-adsorption}}{\overset{\text{dissolution}}{\rightleftharpoons}} AlX^{2+}{}_{(sol)} + AlX_2{}^{+}{}_{(sol)} + AlX_{3\,(sol)} \tag{23}$$

Anions such as $TFSI^-$ and Tf^- tend to adsorb with a high packing density and are reduced by Al^{3+} on the surface. After Al(TFSI) formation, it is desorbed from Al current collector, which leads to the severe corrosion. Once the TSFI product is oxidized, severe corrosion is observed; hence it is not yet used in Li-ion batteries. Al corrosion has been studied by Yang et al.;[84] the corrosion rate with LiTFSI is found to be high compared to other salts, leading to the formation of Al(TFSI). In recent years, various scientific investigations have been reported based on reducing the activity of the $TFSI^-$ ion towards Al. Al_2O_3 has been incorporated as an additive in electrolytes to prevent severe corrosion.[84] In a similar manner HF has also been added to the electrolyte solution (propylene carbonate as a solvent), which drastically suppresses the dissolution of Al, as confirmed by Kanamura et al.[74] In some cases, Al_2O_3 is used as coating substance for the electrodes, thus the coated substance reacts with HF (trace amount in the electrolyte solution) and forms a stable solid–electrolyte interface on the electrode surface; a possible reaction mechanism is give in described below, Equations (24) and (25).

$$Al_2O_3 + 6\,HF \rightarrow 2\,AlF_3 + 3\,H_2O \tag{24}$$

$$Al_2O_3 + 2\,HF \rightarrow 2\,AlOF + H_2O \tag{25}$$

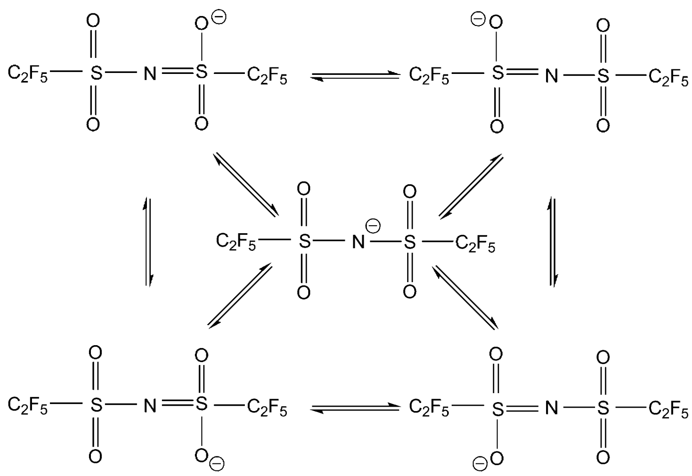
The surface film containingAl_2O_3 is converted to AlF_3 and AlOF; this change in decomposition is independent of the nature of the salt used. It is, however, directly related to the use of HF and the change of composition from Al_2O_3 to AlF_3 and AlOF, which is due to acid–base interaction between the Al_2O_3 and HF. Surface films containing AlF_3 and AlOF are much more stable than the Al_2O_3, so Al corrosion can be suppressed. The addition of HF acid leads to the successful suppression of the corrosion at the Al current collector; however, the use of HF in electrolyte solutions is questionable in practical lithium ion cells.[74]

Lithium Bis(perfluoroethylsulfonyl)imide ($Li[N(C_2F_5SO_2)_2]$ or LiBETI)

$Li[N(C_2F_5SO_2)_2]$ or LiBETI has attracted much attantion in the recent yeas for its superiority and other salient features. This salt was commercialized by the 3M Corporation for research and industrial applications. LiBETI exhibits superior behavior relative to LiTFSI, because of its bulkier anionic size. The larger BETI anion does not form ion-pairs with small cations like Li^+ and Al^{3+}, providing enhanced ionic conductivity. In polymers, the larger BETI anion prevents re-crystallization of the polymer and acts as the solid plasticizer; it improves other requirements like ionic conductivity, electrochemical stability, mechanical stability, and so forth, operating at relatively low-temperatures. Its larger anionic size also favors higher stability of the passivation film; it is less mobile than LiTFSI as well as having a lower viscosity.[85,33] Therefore, the BETI anion should provide the perfect stability of the passivation film without any additives. There are no possibilities of forming HF in non-aqueous solutions and therefore no chance of forming robust LiF on carbonaceous anodes. Further, cells containing LiBETI solution presented a better capacity retention than conventional $LiPF_6$ cells towards carbonaceous anodes.[86] One of the main advantages of using a perfluoinated alkyl group is that owing to the steric shielding of these groups, they have superior stability against the hydrolysis and the possibility of forming HF would be very much reduced. Steric shielding of the nitrogen atom through the hydrophobic perfluorinated alkyl groups and a good delocalization of the negative charge are due to the strong electron-withdrawing effect of these alkyl groups. The delocalization of the negative charge is shown in Scheme 4.

Further, it exhibits a high anodic current, which provides stable passivation layers formed on the both electrode surfaces[86] and also strongly passivates the Al current collector (this property has been improved by the addition of perfluoroethyl groups instead of the trifluoromethyl groups in the LiTFSI salt). The passivation films at the Al current collector contain Al_2O_3 or carbonate layer owing to the bulkier $BETI^-$ ions, which adsorb poorly on Al, and therefore, the reduction of solvent molecules prevails with the surface layer consisting of a dense and protective carbonate species.[83] LiBETI exhibits a wider electrochemical stability window (>5.5 V vs. Li/Li^+) and exhibits good cycling behavior and better thermal stability than LiTFSI and is comparable to that of $LiPF_6$-based electrolytes.[87]

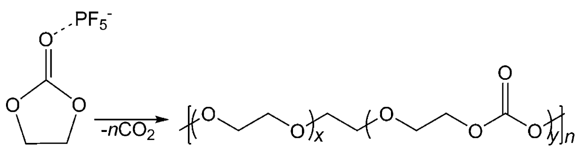
The use of LiBETI is not possible in commercial lithium ion cells owing to the presence of the nitrile group. Users



Scheme 4.

must be cautious, particularly, when using in lithium polymer batteries, owing to the instability of C–F bond and their thermodynamic instability versus metallic lithium.[88] The $BETI^-$ ion has a larger S-N–S bond angle (128.3°) than the $TFSI^-$ ion (109.5°), which leads to asymmetrical bonding with nitrogen atoms in the above-mentioned resonance structures (Scheme 4; the bond angle provides information about the delocalization of charge). However, in $BETI^-$-based systems, delocalization of electron pairs is more complete than for $TFSI^-$ ions.[82]

Moreover, in LiBETI-based solutions, $BETI^-$ ions are not involved in producing polymeric species during the reduction of solvents[89] at the cathode. The polymeric species hinders the release of oxygen and transition-metal atoms in such systems, resulting in much more controlled thermal decomposition of the electrode material. The mechanism of formation of the polymerisation process when using $LiPF_6$ in ethylene carbonate is shown in Scheme 5.[42,90,91]



Scheme 5.

At the cathode, polymeric species are essential for safe battery operation. However, LiBETI-based electrolytes are not able to hinder the release of oxygen atoms; consequently, uncontrolled release of oxygen takes place, especially from oxide-based cathodes, and triggers the thermal runaway of the battery. Thermally stable LiBETI is unable to improve the thermal stability of the electrode, but the thermally unstable $LiPF_6$ can efficiently suppress the decomposition of the cathode and can improve the safety of the cell.[89]

Lithium Tris(perfluoroethyl)trifluorophosphate ($Li[PF_3(CF_3CF_2)_3]$ or LiFAP)

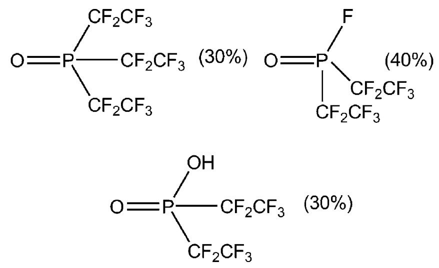
Lithium tris(perfluoroethyl)trifluorophosphate (LiFAP) has been proposed as the potential replacement for commercial cells containing $LiPF_6$-based electrolyte solution.[86] In $LiPF_6$, P–F bonds are labile towards hydrolysis and produce harmful products, as already discussed in previous sections. In LiFAP the fluorine atoms in $LiPF_6$ are partially replaced by perfluoroalkyl groups. Steric shielding of phosphorous through the hydrophobic perfluoroalkyl groups and the delocalization of the negative charge is due to the strong electron-withdrawing nature of the perfluorinated alkyl groups. This leads to superior stability towards the hydrolysis relative to $LiPF_6$ and the bulky perfluorinated alkyl groups weakly coordinate to the Li^+ ions, thus acting as a solid plasticizer to improve the ionic conductivity.[41] The P–F bonds are also stabilised and improve the thermal stability. LiFAP has much a lower reactivity towards carbonaceous anode materials compared to $LiPF_6$, owing to the spatial hindrance of the P–C bonds towards hydrolysis. Gnanaraj et al.[86–93] briefly explained the nature and behaviour of LiFAP salts in a non-aqueous media. The FAP ion is very weakly coordinating bulky anion (anionic size 0.377 nm), leading to a larger ion contact distance; hence the mobility is at moderate level, which leads to lower viscosity and exhibits comparable ionic conductivity with $LiPF_6$.[37] Carbonate species (solvent reduction products) are formed as the main components in solid–electrolyte interface. Lower amounts of HF leads to the lower reactivity of FAP^- ion compared to PF_6^- ions towards the electrodes, which in turn prevents detrimental electrode–solution reactions and favours the development of surface films.[86–99]

LiFAP can be prepared in any lab and the preparation procedure is described in Equation (26).[92]

$$(CF_3CF_2)_3PF_2 + LiF + 5\,H_2O \rightarrow Li[(CF_3CF_2)_3PF_3] \cdot 5\,H_2O \qquad (26)$$

The FAP anion is very stable at room temperature, exhibits a large electrochemical window of approximately 5.0 V versus Li/Li^+ and its anodic current is high; this facts are indicative that a stable solid–electrolyte interface could be formed on carbonaceous anodes and cathodes. Excellent capacity retention is observed compared to other salts tested. LiFAP is thermally stable up to 220 °C; increasing the temperature beyond 220 °C leads to its decomposition and produced following three new products as shown in Scheme 6.

The lithiation and de-lithiation process in LiFAP solution is very sluggish compared to $LiPF_6$, owing to the thin resistive films formed on the surface of the electrodes, particularly carbonaceous anodes. At room temperature, conductivities of approximately 8.2 and 10.7 $mS\,cm^{-1}$ are observed for LiFAP and $LiPF_6$, respectively, in ethylene carbonate/dimethyl carbonate (50:50). Moreover, LiFAP participates in the formation of polymeric species at the cathode.[41] LiFAP exhibits very low irreversible capacity loss (19 %) compared



Scheme 6.

to $LiPF_6$-based electrolytes (49 %), which was measured towards $LiMn_2O_4$ electrodes at 30 °C.[100]

A single-salt system containing LiFAP has higher capacity (330–340 $mAh\,g^{-1}$) retention relative to $LiPF_6$ (310–320 $mAh\,g^{-1}$) at ambient temperature ranges against graphite. A double-salt system containing LiFAP/$LiPF_6$ also showed good capacity retention (300–350 $mAh\,g^{-1}$) even at 80 °C and there no capacity fading is observed. It is believed that LiFAP/$LiPF_6$ binary salt system could be used in the rechargeable batteries at that operate at higher temperatures (up to 80 °C), which is beneficial for electric vehicle applications (Figure 7). In this system, insertion and reinsertion of

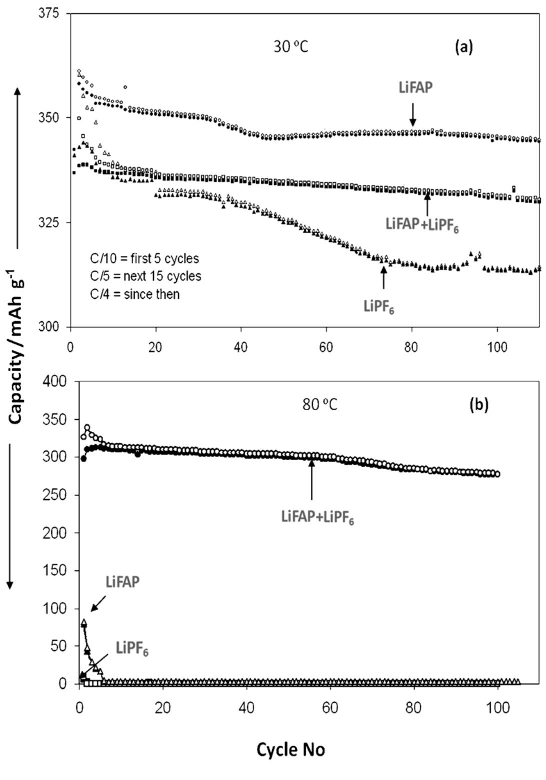


Figure 7. Capacity vs. cycle number of graphite electrodes during galvanostatic lithiation–delithiation, C/4 rates in LiFAP, $LiPF_6$ (1 M) and LiFAP/$LiPF_6$ (0.5 each) solutions in ethylene carbonate/diethyl carbonate/dimethyl carbonate 2:1:2 at 30 and 80 °C, as indicated. The electrodes were cycled in the 0.005–1.6 V domain (vs. Li reference electrodes). The irreversible capacities at 30 °C were usually around 30 % for the solutions containing $LiPF_6$ and around 20–25 % for the LiFAP solution.[91] Reproduced with permission from Elsevier.

Li^+ is very much improved with respect to a single-salt system, because of its compatibility with insertion hosts.[91]

The solid–electrolyte interface contains mostly organic and inorganic carbonates in LiFAP and trace amounts of LiF, whereas for $LiPF_6$ the solid–electrolyte interface contains mostly salt reduction products and alkoxy products.[86] In the case of LiFAP/$LiPF_6$ double-salt system a different type of solid–electrolyte interface was formed due to the presence of small amount of HF; however, in this binary system HF plays a positive effect. Hence, the major reduction products formed are $[PF_2(CF_3CF_2)_4]^-$ and CF_3CH_2H. Owing to the neucleophilic attack of F^- towards the FAP anion and trapping of the protons from HF solutions, a relatively low dissolution process for the Li^+ ion results. Fluorinated compounds in the electrolytes readily to react with lithiated graphite to form very stable and resistive LiF and lithium carbide species on the surfaces, which leads to the stabilization of graphite and improved the capacity retention.[101]

Lithium Fluoroalkylborate ($Li[BF_3(CF_3CF_2)]$ or LiFAB)

In non-aqueous solvent $LiBF_4$ strongly coordinates with Li^+ ions due to the strong coordination ability of BF_4^- ions. This results in ion-pair formation, leading to poor ionic conductivity in aprotic solvents. Some attempts have been made to make supress the BF_4^- ion coordination towards Li^+ ions. By using substituting one of the F atoms with CF_3CF_2, Ue and co-workers[37,102–109] successfully reduced the coordinating ability of the boron anion (FAB) towards the Li^+ ion, which increased the ionic conductivity, even with respect to $LiPF_6$, at <-10 °C in ethylene carbonate/ethylmethyl carbonate (30:70 vol %) (LiFAB ca. 1.8 $mS\,cm^{-1}$; $LiPF_6$ ca. 1 $mS\,cm^{-1}$). When the temperature increases, the conductivity tends to decrease, but remains comparable to $LiPF_6$, and far better than the bare $LiBF_4$ salt. Room-temperature conductivity of LiFAB (ca. 7.8 $mS\,cm^{-1}$) is comparable with that of $LiPF_6$ (ca. 8.5 $mS\,cm^{-1}$) (conductivity was measured in ethylene carbonate: dimethyl carbonate solution with 30:70 ratio).[109]

With respect to BF_4^-, in FAB ions one fluorine atom is replaced by a perfluoroalkyl group (CF_3CF_2), which hinders the strong coordination with Li^+ ion and reduces the ion-pair formation. The ionic mobility is lower in the case of LiFAB relative to $LiBF_4$, as it depends on the size of the anion—FAB^- (0.289 nm) is bulkier than BF_4^- (0.227 nm).[37] Increasing the anionic size leads to the expansion of the ion-contact distance, which in turn leads to a decrease in coordinating ability of Li^+, resulting in enhanced ionic conductivity. The larger size of the anion is beneficial for polymer-based systems, because it acts as solid plasticizer and passivates the Al current collection.[109]

One of the major advantages of using LiFAB is the formation of a passivation film that covers the Al current collector (the passivation film formation is slower than for other salts like $LiBF_4$ and $LiPF_6$). Till now, there is no further information available for the corrosion of Al current collector, but LiFAB appears to be thermally more stable than the parent $LiBF_4$.[109] LiFAB forms a solid–electrolyte interface on both anodes and cathodes. Moreover, this interface is very stable even at elevated temperatures relative to those formed with $LiPF_6$ and $LiBF_4$ salts.[109] LiFAB salt is electrochemically stable up to 4.25 V, so it can be used for 4 V lithium ion cells. Further it is known that inorganic anions are more stable than the organic anions, hence exceeding the electrochemical stability is not advisable. LiFAB and $LiPF_6$ salts deliver the same discharge capacity at 0.2 C (vs. graphite). When increasing the rate from 0.2 to 2 C, the discharge capacity is drastically reduced with respect to $LiPF_6$, but better than $LiBF_4$. [Note: The symbol C denotes the rate capacity of a cell or battery. A battery's capacity is measured in Amp-hours, called "C". The theoretical amount of current a battery delivers when discharged in one hour is 100%. For example, a battery with 1000 mAh can submit a current of 1000 mA for 1 h, when the rate is 1 C. If the same battery is discharged at 0.5 C, the discharge rate is 500 mA for 2 h. At 2C, a 1000 mAh battery delivers 2000 mA for 30 min.] Further, capacity fading is noted to be a function of cycle number. LiFAB demonstrated superior performance than $LiPF_6$- and $LiBF_4$-based electrolytes against a graphite anode. In addition, HF contamination in LiFAB salts is found to be much less[109] relative to $LiPF_6$ and $LiBF_4$.[110,111]

Lithium Tris(trifluoromethanesulfonyl)methide ($Li[C(CF_3SO_2)_3]$ or LiTFSM)

Dominey[112,113] from Covalent Associates (USA) successfully synthesized this lithium salt based on a carbanion stabilized by three perfluorinated trifluoromethanesulfonyl groups, nick-named "Li methide". The anion is very stable due to steric hindrance that exists in the pyramidal geometry of the methide anion. LiTFSM exhibits better ionic conductivity than LiTFSI. This behaviour is due to its wide range of dissolution in various non-aqueous solvents, which is attributed the effective delocalization of the anion, leading to higher ionic conductivity.

Strong Li–C bonds are involved in the formation of surface films on both electrode surfaces.[114] The following mechanism [Eqs. (27)–(29)] does not describe the complete reduction process of TFSM anion, because it has several functional groups (e.g., C–S, S–O and C–F). The C–F bonds in the CF_3 groups strongly react with Li^+ and formed LiF species. Similarly SO_2 reacts with Li^+ and to form compounds like $Li_2S_2O_4$ and Li_2S.[115] The mechanism shown in Equations (27)–(29) gives information about the reduction process[79] of LiTFSM.

$$Li[C(SO_2CF_3)_3] + n\,e^- + n Li^+ \rightarrow 0.5\,Li_2S_2O_4 + m\,LiF + 0.5\,C_2F_xLi_z + Li_2[C(SO_2CF_3)_2] \quad (27)$$

$$Li[C(SO_2CF_3)_3] + 2\,e^- + 2\,Li^+ \rightarrow Li_2[C(SO_2CF_3)_2] + Li(CF_3SO_2) + \text{etc.} \quad (28)$$

$$Li[C(SO_2CF_3)_3] + 4\,e^- + 4\,Li^+ \rightarrow Li_2SO_3 + Li_2S + Li_2O \quad (29)$$

LiTFSM is stable up to 340 °C and it could be used for high-temperature operations. Besides, LiTFSM has a good oxidation potential (6.1 V), which was confirmed by Ue et al.[52] using HOMO energy calculations (V vs. Li/Li^+). When compared to inorganic lithium salts, LiTFSM demonstrated slightly lower oxidation potential: $[CF_3SO_3]^-$ (5.9 V) < ClO_4^- (6.0 V) < $[N(CF_3SO_2)_2]^- \cong [C(CF_3SO_2)_3]^-$ (6.1 V) < BF_4^- (6.2 V) < PF_6^- (6.3 V) < AsF_6^- (6.5 V).

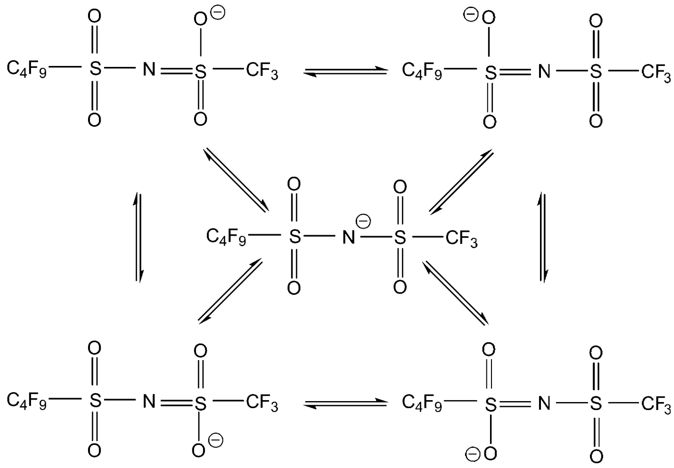
In earlier days it was reported that, LiTFSM is inert at the Al current collector;[45] however, recently information to the contrary was reported.[84] LiTFSM can strongly react with Al, leading to severe corrosion. Nonetheless, the corrosive behaviour is slightly lower than LiTFSI. During the reaction with Al, a large amount of $Al[C(CF_3SO_2)_3)]$ is formed in the passivation film,[84] which is unable to protect the Al against corrosion. Severe corrosion is observed beyond 4.5 V and it is noted that the potential use of LiTFSM in high-voltage cathodes is questionable, because Al is universally adopted as a current collector.

Lithium Nonafluorobutylsulfonyltrifluoromethylsulfonylimide ($Li[N(C_4F_9SO_2)(CF_3SO_2)]$ or LiFBMSI)

A new asymmetric type of imide salt, lithium nonafluorobutylsulfonyl trifluoromethylsulfonyl imide (LiFBMSI has been reported in recent years as prospective candidate for $LiPF_6$ replacement.[76] LiFBMSI exhibits some superior properties with respect to its counterparts discussed in the previous sections. The presence of the perfluoroethyl groups (with strong electron-withdrawing properties) leads to a strong hydrophobic nature towards hydrolysis. Increasing the length of the fluoroalkyl groups results in strong passivation phenomena at the Al current collector and enhancement of conductivity.[76] LiFBMSI exhibits an ionic conductivity of approximately 3.5 $mS\,cm^{-1}$ in propylene carbonate/DME (1:2) solution and showed high oxidation potential values 5.9 V vs. Li/Li^+, due to the presence of the bulkier anion. It is evident that the oxidation potential is increased by increasing the length of the fluoroalkyl groups: $Li[N(CF_3SO_2)_2]$ (5.2 V) < $Li[N(C_4F_9SO_2)(CF_3SO_2)]$ (5.9 V) < $Li[N(C_8F_{17}SO_2)(CF_3SO_2)]$ (6.0 V)

A comparison of ionic conductivity values with other salts at room temperature conditions (0.5 M LiFBMSI in propylene carbonate/DME (1:1)[83]) is given here: $Li(CF_3SO_2)$ < $Li[N(C_2F_5SO_2)_2]$ < $Li[N(C_4F_9SO_2)(CF_3SO_2)]$ < $Li[N(CF_3SO_2)_2]$ < $LiPF_6$

If the chain length is increased to C_8F_{17} in the imide salts, a drop in the conductivity results. This is due to bulkier anionic size, which leads lower mobility. However, LiFBMSI salt showed good cycling efficiency with respect to other salts discussed: $LiPF_6 < Li[N(C_2F_5SO_2)_2] < Li[N(C_4F_9SO_2)(CF_3SO_2)]$. Kanamura et al.[74] have studied passivassion phenomena on Al current collectors using XPS and FT-IR spectroscopy. The XPS study confirmed the presence of Al_2O_3 and AlF_3 species, though the amount of latter is found to be very small. Al_2O_3 is the main component on the surface film and the AlF_3 component amounts to 10–20 % of the solid–electrolyte interface. The XPS study also confirmed that surface films were rarely changed during anodic polarization and $FBMSI^-$ species were also observed on the surface films. The strong delocalization of negative charge of the FBMSI anion is shown in Scheme 7.



Scheme 7.

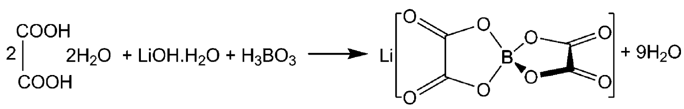
LiFBMSI is electrochemically stable up to 5.0 V vs. Li/Li^+ in non-aqueous solvents. During the anodic polarization of Al, no peaks were observed in FTIR studies and it confirmed the stabilization of Al up to 5.0 V against this LiFBMSI-based electrolyte.[116] Cyclic voltammetry studies have also been performed and confirmed the stability window up to 4.8 V vs. Li/Li^+; beyond 4.8 V very slow corrosion of Al was noted. Improvements of Al suppression were also reported by the same author; while introducing HF into the electrolyte solution, the surface film is changed from Al_2O_3 to AlF_3–Al_2O_3 (a similar kind of mechanism was already described for LiTFSI). Therefore, the function of LiFBMSI is related to the physiochemical properties of imide salts and the $FBMSI^-$ ion does not form a contact ion pair with small cations like Li^+ and Al^+. LiFBMSI-based electrolytes exhibit good passivation phenomena towards the Al current collector without any additives, which indicates that it is one of the best candidates to replace $LiPF_6$ in commercially available batteries.

Lithium Bis(oxalato)borate ($Li[B(C_2O_4)_2]$ or LiBOB)

Recently lithium bis(oxalato)borate LiBOB has attracted much attention because of its peculiar properties.[117–136] It has been found that BOB anions are actively involved in the solid–electrolyte interface formation on carbonaceous anodes as well as cathodes, and this kind of behaviour is not observed for other salts. The synthesis procedure of this salt is quite easy and there are no possibilities of forming the dangerous by-products like hydrofluoric acid (HF). A few of the peculiar properties of this salt are listed below.

1) The crystalline state of LiBOB is more stable than that of other salts like $LiPF_6$, and so forth.
2) It has the ability to form a stable solid–electrolyte interface.
3) It has good stability in a wide potential window.
4) It shows acceptable solubility in alkyl carbonate organic solvents like ethylene carbonate, dimethyl carbonate, diethyl carbonate, and so forth.
5) It exhibits high conductivity in various non-aqueous solvent systems.
6) It shows good cycling behavior.
7) It has excellent overcharge tolerance.
8) It is environmentally benign.
9) It exhibits better thermal stability than $LiPF_6$ in organic solvents.[67,117,118]

LiBOB can be prepared by simple and known technique (Scheme 8), that is, a solid-state reaction method, and this route yields the mass production of LiBOB, which reduces

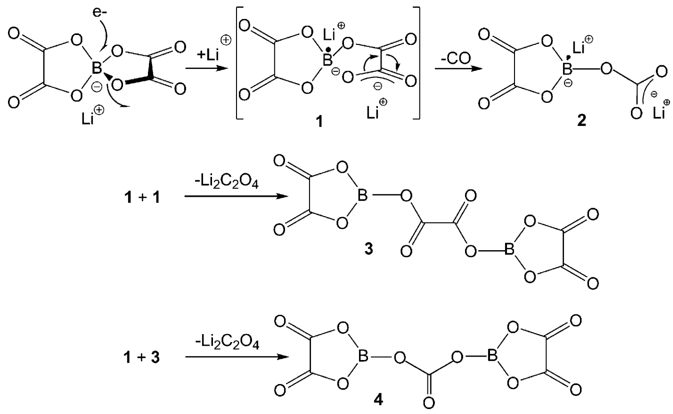


Scheme 8.

the cost of the battery. Angell et al.[117] and Lishka et al.[119] first reported the preparation of LiBOB and followed by Yu et al.[120] Recently, Xu and Angell suggested[122] that highest ambient temperature conductivity could be achieved for LiBOB in acetonitrile.

The BOB anion is effectively involved the formation of solid–electrolyte interface, which mostly contains semicarbonates. These semicarbonates are able to prevent the solvent molecules from undergoing co-intercalation into the carbonaceous anodes. LiBOB has been tested with or without ethylene carbonate (as co-solvent); the stability of solid–electrolyte interface improves notably in the presence of ethylene carbonate.[123] It is apparent that this kind of solid–electrolyte interface can prevent the solvent molecule co-intercalation and it even stabilizes the graphite anode when exfoliating propylene carbonate is used as a co-solvent.[125] The decomposition mechanism of LiBOB in the presence of ethylene carbonate is given in Scheme 9.

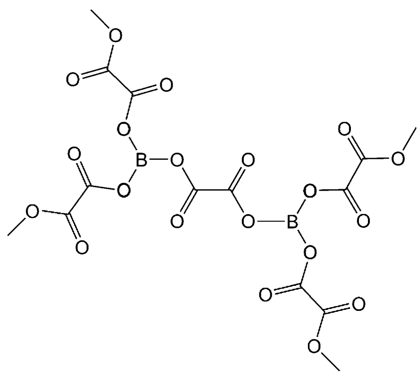
The above carbonyl-rich species form the main components in the solid–electrolyte interface and hinder the solvent co-intercalation during the cycling process. In addition



Scheme 9.

to the semicarbonates, some other species are also formed in the solid–electrolyte interface; these include inorganic species, such as $Li_2C_2O_4$, $LiBO_2$ and Li_2CO_3, and polymeric moieties, like polyether, polyalkane and polycarbonates (which are formed during from solvent molecules through ethylene carbonate and dimethyl carbonate reduction, for example). So multiple mechanisms are also possible and the end product will contain a high content of carbonyl groups, which form the semicarbonates. Hence, LiBOB initiates the polymerisation process that leads to the formation of oligomeric borates with a carbonyl-rich phase as shown in Scheme 10.

BOB is a bulky anion that favours delocalization of charge, reduced anion-to-cation reaction and low-lattice energy, and leads the high degree of dissociation of the salt. These bulky anions trap the amorphous regions at ambient temperatures when using polymeric systems, and show better conductivity than $LiBF_4$ and comparable conductivity to commercially used $LiPF_6$. At elevated temperatures, the



Scheme 10.

conductivity is in the order of 10^{-3}–10^{-4} $S\,cm^{-1}$.[126] Improved capacity retention at high temperatures was achieved by researchers interested in electric vehicle applications that also require rechargeable lithium batteries[126] without HF formation. Solid composite polymer electrolytes are better candidates for future lithium rechargeable batteries. By adding nanoparticles with high surface area to the above polymer electrolyte, an increase in the conductivity and other properties mentioned above are observed. Gel electrolyte systems at elevated temperature (60–70 °C) exhibit a slightly higher capacity than at room temperature[135] (Figure 8).

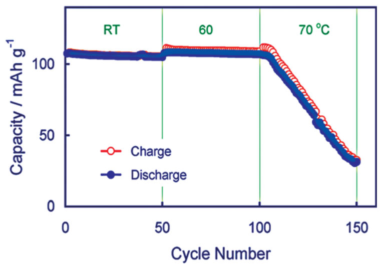
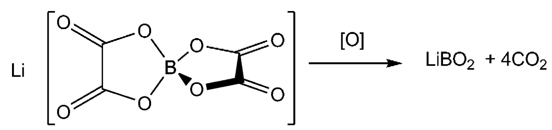


Figure 8. Cycling performance of a liquid electrolyte $Li/LiMn_2O_4$ cell using 1.0 M LiBOB 1:1 propylene carbonate/ethylene carbonate electrolyte at different temperatures. The cell was cycled at a constant current rate of 0.5 C between 3.5 and 4.2 V. From reference [135]; reproduced with permission from Elsevier.

Use of LiBOB as the electrolyte salt might prevent the dendrite formation on the Li metal anode when propylene carbonate is used as a plasticizer. When platinum is used as working electrode, electrochemical stability is observed up to 4.5 V. Further, a high anodic current is present, indicative of stable solid–electrolyte interface formation on both anode and cathode surfaces,[136] which also protects the Al current collector from corrosion.[137,138]

During overcharge, LiBOB produced only a small amount of smoke as long as the temperature did not exceed 100 °C and it did not catch fire or spark, whereas $LiPF_6$-based electrolytes not only caught fire, but resulted in a violent explosion and the maximum temperature reaching up to 400 °C. This type of tolerance is attributed to the oxalate molecular moieties of LiBOB, which are preferably oxidized and produce CO_2 through the release of oxygen from the cathode.[139] The possible decomposition mechanism of over charge tolerance is shown in Scheme 11.

During mild oxidation of LiBOB, the internal pressure is rapidly built up by the released CO_2, which immediately opens a "safety vent" before thermal runaway or explosion occurs.[139,140] Adding small amount of LiBOB to $LiBF_4$

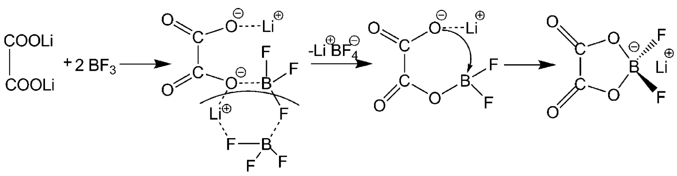


Scheme 11.

showed enhanced cycling performance in propylene carbonate based electrolytes.[140–144] Similarly, when LiBOB is added to $LiPF_6$, the binary mixture delivers an appreciable improvement in the capacity retention compared to $LiPF_6$ alone in a $LiMn_{1/3}Ni_{1/3}Co_{1/3}O_2$/mesocarbon microbeads cell. Nevertheless, the discharge capacity is slightly lowered in the binary mixtures relative to $LiPF_6$. It is interesting to note that pure LiBOB electrolyte solution exhibited very stable discharge behaviour with lower capacity values than the binary mixture or $LiPF_6$ alone in the above cell at elevated temperatures.[145] A synergistic effect is observed when LiBOB is used as a mixed salt with either $LiPF_6$ or $LiBF_4$. The cell performance, based on measures such as discharge capacity and capacity retention of $LiPF_6$/LiBOB or $LiBF_4$/LiBOB mixtures, after long cycling at room temperature is much higher than that of each single salt. It is apparent that the salt blend is a good way to achieve superior cell performance.[146–150]

Lithium Oxalyldifluoroborate ($Li[BF_2(CO_2)_2]$ or LiODFB)

To overcome the following setbacks of the LiBOB salt, a new lithium oxalyldifluoroborate (LiODFB) has been introduced as alternative. LiBOB exhibits limited solubility in linear carbonates, it is extremely sensitive to impurities, such as the trace amount of unreacted reactants, has a high viscosity in the electrolyte solutions and a highly resistive nature of solid–electrolyte interface, which reduces the power and capability of the cell. In addition to that, poor cell performance at lower temperature, high onset temperature and formation of gaseous byproducts led to a search new salts in place of LiBOB.[150] In 2007, Zhang[139,151] proposed a new salt, LiODFB, to circumvent the aforementioned demerits of the LiBOB-based electrolyte. In LiODFB one of the oxalate groups from LiBOB is replaced by fluorine and the synthetic route is shown in Scheme 12.[152–158]



Scheme 12.

LiODFB is extremely soluble in linear carbonates, which lead to lower viscosity and better ionic conductivity than found with LiBOB at low temperatures. In addition to the said advantages of LiBOB, LiODFB has some salient features that are listed below.

1) LiODFB effectively passivates the Al current collector even at high potentials.
2) A stable solid–electrolyte interface is formed under elevated temperatures.
3) It exhibits excellent over-charge tolerance that increases the safety protection of the battery.
4) It has a very high cycling efficiency.

For comparison purposes, the ionic conductivity has been measured for different salts (electrolyte solution containing propylene carbonate/ethylene carbonate/ethylmethyl carbonate 1:1:3) at low temperature, below $-30\,°C$. The conductivity values follow the order $LiBF_4 \approx$ LiODFB $>$ LiBOB. Increasing the temperature, the conductivity lies between $LiBF_4$ and LiBOB, owing to the combined salt dissociation and viscosity. The improved electrolytic conductivity is due to the presence of both $LiBF_4$ and LiBOB moieties.[139] The electrolytic conductivity is better than $LiBF_4$ and comparable to $LiPF_6$. In addition Al is strongly passivated and the oxidation of the electrolyte solution is also suppressed. Moreover wider electrochemical stability (>5 V) shows that it could be used in high potential cells. The chemical combination of Al^{3+} and B−O molecular moieties forms a dense protecting layer on the Al surface. Recently, Li et al.[156] and Zhang et al.[157] reported conductivity studies on the LiODFB system in various combinations of solvents. Among them, ethylene carbonate/dimethyl carbonate (1:1 v/v) combination exhibited superior performance—approximately 8.58 mS cm^{-1} under ambient temperature conditions (Figure 9).

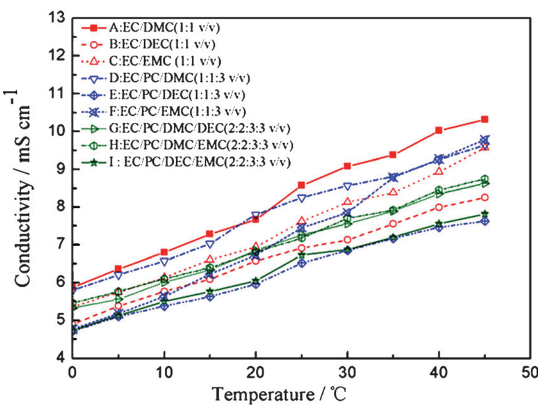
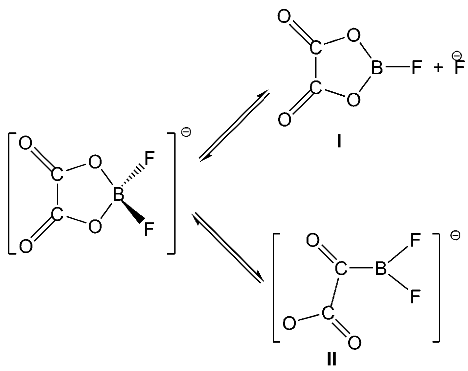


Figure 9. Conductivities of 1 M LiODFB salt in various solvent systems (0–45 °C);[156] reproduced with permission from Elsevier. EC = ethylene carbonate, DMC = dimethyl carbonate, DEC = diethyl carbonate, EMC = ethylmethyl carbonate, PC = propylene carbonate.

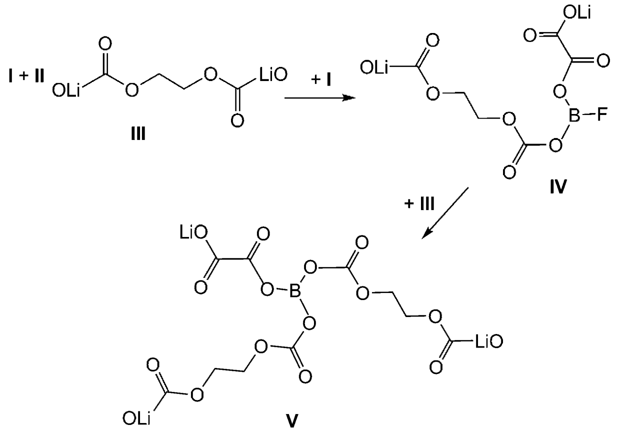
Like LiBOB, LiODFB is involved in the formation of the solid–electrolyte interface, leading to a semicarbonate structure, by means of two simultaneous chemical equilibria (Scheme 13).

Moieties **I** and **II** can combine with the main solid–electrolyte interface components, such as lithium semi-carbonates (**III**) to form more complicated and stable oligomers (Scheme 14).

Evidently, these reactions do not involve any electronic transfer and a similar mechanism is observed in electrolytic additives such as vinylene carbonate. The reaction leading



Scheme 13.



Scheme 14.

to **II** occurred owing to the presence of trace amounts of oxalate moieties (−CO−COOLi). Relative to LiBOB, LiODFB has a much shorter voltage plateau, which is due to the fact that it only has one oxalate group and while the former has two. LiODFB also exhibits excellent stabilization even at high concentrations of propylene carbonate (50%), and provides excellent high- and low-temperature performances, which is due to the relatively low charge-transfer resistance of the cell. LiODFB is stable up to approximately 240°C (lower than LiBOB at ca. 306°C); this low decomposition temperature is very useful for safety in an "abuse" environment, that is, over charge and short circuiting. In such "abuse" conditions, enormous amount of heat are generated that cause thermal runaway of the battery; if a safety vent cannot open immediately, an explosion of battery results. LiODFB generates gaseous products, like LiBOB, and therefore makes it possible to open the safety vent before thermal runaway occurs.[139] Monikowska et al.[152] reported that a binary mixture of $LiBF_4$ and LiODFB in ethylene carbonate/propylene carbonate/dimethyl carbonate (1:1:3) exhibited the conductivity values around 10^{-3} $S\,cm^{-1}$ at room temperature. The binary mixture with $LiPF_6$ also was reported, which resulted enhancement of ionic conductivity, and excellent cycling profiles were noted at room temperature and at elevated temperatures. The cycling profiles of $LiFePO_4$/AG (AG = artificial graphite) with different molar ratios of $LiPF_6$ demonstrated that at high concentrations of $LiPF_6$ (50%) a poor performance of the cell is observed.[158] This result proves the possibility of using mixed-salt-based electrolyte solutions in practical lithium ion cells.

Other Salts

In Table 1 we have summarized some of the salts studied as possible electrolyte salts for lithium batteries by several researcher groups.[37,76,110,111,112,136,139] Most of the salts contain hydrocarbons, which severely affect the solid–electrolyte interface as they are involved in the robust interface formation. Hence, the solid–electrolyte interface contains the multiple components including protons, which also exfoliate the carbonaceous anodes causing a breakdown of interface and the possibly leading to thermal runaway.[158] Most of the studied salts were bulkier anions and it is believed that such anions strongly passivate the Al current collector. Moreover, the anions also act as solid plasticizers (when using a polymer system) and thereby enhance the ionic conductivity of the system.

$Li(C_nF_{2n+1}SO_3)$ (n = 1, 4 and 8): In this series of electrolytes, moderate ionic conductivity is exhibited in solution when $n = 1$, that is, LiTf as already described in previous sections. Increasing the length of the fluoroalkyl group's chain from CF_3 to C_4F_9 resulted in no evident difference in the conductivity values. However, increasing the chain length from CF_3 to C_8F_{17} showed a decrease in conductivity due to the poor mobility of the anions in the solution (bulkier anionic size restricts the formation of contact ion pairs; conductivity: $Li(CF_3SO_3) = LiC_4F_9SO_3 > LiC_8F_{17}SO_3$), whereas oxidation potential increased in the following order $Li(CF_3SO_3) < LiC_4F_9SO_3 < LiC_8F_{17}SO_3$.

$Li[N(ROSO_2)_2]$ (R = CF_3CH_2, $HCF_2CF_2CH_2$, $CF_3CF_2CH_2$ and $(CF_3)_2CHOSO_2$): The imide ester salts showed better cycling performance than other salts discussed and they are promising salts for Li-ion cells; however, they are questionable in practical cells due to the presence of hydrocarbons.[76] Imide ester salts exhibit better performance than imide salts, but the use of them in practical batteries is found to be very difficult, because of the presence of multi-component ionic conductors like protons and Li^+. Hence, there is a possibility of exfoliating the carbonaceous anodes and the formation of a robust solid–electrolyte interface on both electrodes, which hinders Li^+ transport during cycling. Cycling performances of the salts are as follows:[76] $Li[N\{(CF_3)_2CHOSO_2\}_2] > Li[N(C_4F_9SO_2)(CF_3SO_2)] > Li[N(C_2F_5SO_2)_2] > LiPF_6$. Conductivity values of $Li[N(ROSO_2)_2]$ groups are in the order $Li[N\{(CF_3)_2CHOSO_2\}_2] > Li[N(CF_3CH_2OSO_2)_2] = Li[N(CF_3CF_2CH_2OSO_2)_2] > Li[N(HCF_2CF_2CH_2OSO_2)_2]$ and the oxidation potentials are as follows $Li[N\{(CF_3)_2CHOSO_2\}_2] > Li[N(CF_3CF_2CH_2OSO_2)_2] > Li[N(HCF_2CF_2CH_2OSO_2)_2] > Li[N(CF_3CH_2OSO_2)_2]$.

Table 1. Salts studied as possible electrolyte salts for lithium batteries.

Salts	Conductivity [mS cm^{-1}] at 25 °C						Oxidation potential [V][b]	Ref.
	PC	EC/DMC (1:1)	EC/DMC/DEC (2:2:1)	PC/DME (1:2)	DME	DMSO		
$Li(CH_3BF_3)$		3.4						[110]
$Li(CF_3BF_3)$	5.3	6.1					5.2	[110,111]
$Li(CF_3CF_2BF_3)$	5.3						5.3	[111]
$Li(n\text{–}C_3F_7BF_3)$	4.5						5.5	[111]
$Li(n\text{–}C_4F_9BF_3)$	4.1						5.7	[111]
$Li[(CF_3)_2BF_2]$		7.2						[110]
$Li[(CF_3)_4B]$			10.1					[37]
$LiCF_3CO_2$				0.4				[76]
$LiN(CF_3CO_2)_2$				0.8				[76]
$LiCH_3SO_3$				insoluble				[76]
$LiC_4F_9SO_3$				2.3			6.0	[76]
$LiC_6F_5SO_3$				1.1				[76]
$LiC_6H_5SO_3$				0.2				[76]
$LiC_8F_{17}SO_3$				1.9				[76]
$LiN(FSO_2C_6F_4)(CF_3SO_2)$				3.0			6.5	[76]
$LiN(C_8F_{17}SO_2)(CF_3SO_2)$				3.2			6.0	[76]
$LiN(CF_3CH_2OSO_2)_2$				3.0			5.4	[76]
$LiN(CF_3CF_2CH_2OSO_2)_2$				3.0			5.6	[76]
$LiN(HCF_2CF_2CH_2OSO_2)_2$				2.9			5.5	[76]
$LiN((CF_3)_2CHOSO_2)_2$				3.1			5.8	[76]
$LiC(CF_3CH_2OSO_2)_3$				2.9				[76]
$LiB[OC(CF_3)_2]_4$					11.1	4.86	4.5	[136,139]
$LiB(C_3H_2O_4)_2$	3.92					4.99		[139]
$LiPF_5(CF_3)$				4.2			6.2	[160]
$LiPF_4(CF_3)_2$				3.9			6.2	[160]
$LiPF_3(CF_3)_3$				3.9			5.1	[160]

[a] PC = propylene carbonate, EC = ethylene carbonate, DEC = diethyl carbonate, DMC = dimethyl carbonate, DME = dimethyl ether, DMSO = dimethyl sulfoxide. [b] Pt electrode in PC (V vs. Li/Li$^+$).

$Li[BF_{4-n}(CF_3)_n]$ (n = 1, 2, 3 and 4): Increasing the size of the anion leads to weaker in coordination with Li^+ ion, resulting in higher dissociation that provides enhanced conductivity and decrease in ionic mobility, which is governed by free anions and depends on size of the anion.[159–164] The conductivities of $Li[BF_{4-n}(CF_3)_n]$ (n = 1, 2, 3 and 4) are as follows: $LiBF_4 < Li[BF_3(CF_3)] < Li[BF_2(CF_3)_2] < Li[B(CF_3)_4]$

The extraordinary conductivity of $Li[B(CF_3)_4]$ is attributed to the extremely high dissociation that comes from the delocalizing or shielding effect of negative charge of the boron atom by CF_3. The conductivity trend of $Li[BF_{4-n}(CF_3)_n]$ is different from $Li[PF_{6-n}(CF_3)_n]$, because the latter are large enough to dissociate while introducing the perfluoroalkyl groups, which leads to a slight decrease in conductivity compared to parent $LiPF_6$. In this group, the use of C_2F_5 instead of CF_3 (i.e., $Li[BF_3(CF_3CF_2)_3]$) leads to a better performance than rest of the members in this group, which has been already discussed detail in previous section. So other salts have some academic interests, but they are not fit for their use in practical cells.

$Li[PF_{6-n}(CF_3)_n]$ (n = 1,2 and 3): In this context the hydrophobically unstable P–F bonds of PF_6^- are replaced with P–CF_3 bonds of the $[PF_{6-n}(CF_3)_n]^-$ ions.[160] They coordinate very weakly with Li^+ ions. By increasing the value of n, the dissociation increases, but the conductivity is not drastically decreased; however, a slight variation can be observed due to the different sizes of the anions. At the same time oxidation potential slightly increases in the following manner: $Li[PF_5(CF_3)] = Li[PF_4(CF_3)_2] > LiPF_6 > Li[PF_3(CF_3)_3]$.

Among the four types of salts analysed, $Li[PF_4(CF_3)_2]$ shows the best performance and the cycling profile is equal to that of $LiPF_6$; however, ionic conductivity values are varied as follows: $LiPF_6 > Li[PF_5(CF_3)] > Li[PF_4(CF_3)_2] > Li[PF_3(CF_3)_3]$.

Solution impedance measurements were recorded for storage; a high value of impedance is observed in $LiPF_6$-based electrolytes and only a very small increase of the impedance values was found in a $Li[PF_4(CF_3)_2]$ containing electrolyte.

Thermal stability of anions follows the order $[PF_4(CF_3)_2]^- > [PF_5(CF_3)^-] > [PF_3(CF_3)_3^-] > PF_6^-$ and order of ion dissociation ability is $Li[PF_3(CF_3)_3] > Li[PF_4(CF_3)_2] > Li[PF_5(CF_3)] > LiPF_6$.[160]

Surface analyses were also performed by using XPS; at low salt ($Li[PF_4(CF_3)_2]$) concentration the conductivity of surface layer was very poor, which reflected in overall cell performance.

$Li[N(RSO_2)(R'SO_2)]$ (R = C_6F_5, C_4F_9 and C_8F_{17}; R′ = CF_3): This series, containing much longer alkyl groups, favours good ionic conductivity and excellent passivation towards the Al current collector. $Li[N(C_4F_9SO_2)(CF_3SO_2)]$ salts showed the better conductivity (ca. 3.5 mS cm^{-1}) among the group and its characteristics were explained in a previous section.[76,161–163] The conductivity values are found to be in the following order: $Li[N(C_4F_9SO_2)(CF_3SO_2)] > Li[N(C_8F_{17}SO_2)(CF_3SO_2)] > Li[N(C_6F_5SO_2)(CF_3SO_2)]$.

Imide salts that have larger alkyl groups do not show higher conductivity, owing to the size of the anions. The larger anionic size leads to the higher viscosity and lower mobility, and there is no possibility of forming ion pairs in the non-aqueous medium. However, they are promising candidates and give better corrosion protection than LiTFSI, so further studies are required to establish the candidates for their use in practical cells.

$Li[N(FSO_2)(n\text{-}C_4F_9SO_2)]$ (LiFNFSI): Based on the previous work by Kanamura et al.,[74] Han et al.[165] followed and reported the synthesis and electrochemical performance of lithium (fluorosulfonyl)(nonafluorobutanesulfonyl)imide (LiFNFSI). Generally, the fluorosulfonyl (FSO_2^-) group in the imide anion causes a decrease in thermal stability. Here, the new salt also experiences poor thermal stability; however, LiFNFSI (179 °C) is much better than commercial $LiPF_6$ (107 °C). The ionic conductivity has been measured for 1 M LiFNFSI in ethylene carbonate/ethylmethyl carbonate (3:7 v/v) solution and compared with $LiPF_6$, $LiClO_4$, and $LiBF_4$ salts in the same conditions. The specific ionic conductivities were observed in the following order: $LiPF_6 > LiFNFSI \approx LiClO_4 > LiBF_4$.[165] Electrochemical stability of 1 M LiFNFSI in ethylene carbonate/ethylmethyl carbonate has been reported on the Pt electrode against Li/Li^+ and found that $FNFSI^-$ ions are stable up to 5.7 V.[165] Al corrosion behavior was also studied and the CV traces were very similar to the non-corrosive salts like $LiPF_6$ and $Li[N(CF_3SO_2)(n\text{-}C_4F_9SO_2)]$. From the CV studies, it could be seen that fluorosulfonyl (FSO_2^-) group in LiFNFSI could be beneficial for forming an effective protective film to prevent the corrosion of Al substrate, while using 4 V class cathode materials.[165] Cycling performance of the 1 M LiFNFSI in ethylene carbonate/ethylmethyl carbonate solution was studied in $LiCoO_2$/graphite and also compared with $LiPF_6$ in various temperatures.[165] At 25 °C, the cell containing the LiFNFSI salt delivered and sustained more stable discharge capacity than that with $LiPF_6$ upon cycling to 100 runs. The total capacity-fade rate was estimated after 100 cycles. The LiFNFSI-based cell experiences only 11 % fade, whereas for $LiPF_6$-based cells a 23 % fade is observed. The capacities for the cell with $LiPF_6$ fade abruptly, in comparison with LiFNFSI, upon the temperature being raised to 60 °C. It seems that the better thermal stability and lower content of HF for LiFNFSI plays an important role in improving the high-temperature resilience of lithium ion batteries.

$Li[BF_3Cl]$: With the idea of increasing the anionic size of $LiBF_4$, the replacement of one of the F atoms by other halogens has been investigated, leading to an asymmetric structure of the salt.[166] The asymmetric salt $Li[BF_3Cl]$ can be prepared by replacing one fluorine atom in $LiBF_4$ with a chloride atom from ammonium chloride. $Li[BF_3Cl]$ is expected to have higher solubility in organic solvents than parent $LiBF_4$. More importantly, the weaker coordination of Cl than F to the central boron atom promotes the formation of BF_3, based on the chemical equilibrium of "$BF_3Cl^- \leftrightarrow BF_3 + Cl^-$" and can then participate in the solid–electrolyte interface formation. The ionic conductivity of $Li[BF_3Cl]$ is almost the same as that of the parent $LiBF_4$. However, at −50 and 50 °C the salt exhibits superior performance than parent salt in ethylene carbonate/ethylmethyl carbonate (3:7) solution. Further, it is involved in the formation of solid–electrolyte interface on graphite and forms a protective layer towards the Al current collector. The BF_3Cl^--based ammonium ions are expected to be superior to the BF_4^- analogues, for application in the electrolytes of double-layer capacitors, especially for low-temperature electrolytes, since the asymmetry of the BF_3Cl^- ion with respect to the BF_4^- ion favours an increase in the dissociation and solubility of the ammonium ion in organic solvents.[165]

Outlook

The state-of-art of Li-ion cell, high-voltage cathode materials, $LiCoPO_4$, $LiNiPO_4$ and so forth have not yet fully explored. This effort is further hampered by the limitation of the decomposition potential of the electrolytes. Extending the stability of the electrolytes is desperately needed when using high-voltage cathode materials. At the same time, safety in operation of the cell and solid–electrolyte interface formation with Al protection are necessary and cannot be compromised. Latest research indicates that using solid–electrolyte interface-forming additives, like vinylene carbonate, is encouraging; nevertheless, it needs further investigation. Results from mixed salts solution are also quite impressive, for example, $LiFAP/LiPF_6$, $LiBOB/LiPF_6$ and $LiBF_4$/LiODFB; however, further studies are required to understand the mechanism. Using green solutions like ionic liquids as electrolytes are appreciated for the eco-friendly environment. However, the same lithium salts (LiBOB and LiODFB) could be used for the Li^+ ion source, so the new salt's anion should be able to form solid–electrolyte interface, like LiBOB does. The setback of using the LiBOB is its limited solubility in the carbonate solvents and it has less decomposition potential than $LiPF_6$. In spite of that, Chemetall has commercialized the product for research and development. Another prospective candidate is LiFAP, very recently commercialised by Merck KGaA, which exhibits superior performance with respect to commercially available $LiPF_6$. Research towards the development of new salts for the replacement of $LiPF_6$ to reduce the battery hazards and improved safeties are warranted. The commercialization of LiBOB and LiFAP opens the new avenues of research for the lithium battery electrolyte as well as high voltage cathodes.

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